

- Q.29 Define screen analysis and explain Tyler standard screen series.
- Q.30 Explain screen capacity and screen effectiveness.
- Q.31 Classify filtration and explain any one type of filter medium.
- Q.32 Explain the working of a filter press.
- Q.33 Describe gravity classifiers with neat sketch.
- Q.34 Explain working of cyclone separator in detail with neat diagram
- Q.35 Describe construction and working of a two-arm kneader.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the principle, construction and working with advantages of rotary drum filter with neat and clean diagram.
- Q.37 Describe the principle construction and working of plate and frame filter press with neat and clean diagram.
- Q.38 Write short notes on any two.
- Grizzlies
 - Tyler standard screen series
 - Stationary screen

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**3rd Sem / Chem, P & P, Chem Engg. (Spl. Paint Tech.),
Chem Engg. (Spl. Polymer Engg.)
Sub : Mechanical Operations**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following best represents particle shape?
- Sphericity
 - Porosity
 - Void fraction
 - Density
- Q.2 Which equipment is used for fine grinding?
- Jaw crusher
 - Ball mill
 - Roll crusher
 - gyratory crusher
- Q.3 Which filter operates under vacuum?
- Filter press
 - Leaf filter
 - Rotary drum filter
 - Plate and frame filter
- Q.4 Which of the following works on the principle of impact?
- Gyratory mill
 - Jaw crusher
 - Ball mill
 - Roll crusher
- Q.5 Sphericity of a special particle is _____
- Zero
 - One
 - Infinite
 - None

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- Q.6 The fluid energy mill involves _____
- Pressure
 - Temperature
 - Fluid
 - Density
- Q.7 Which of the following is a continuous filter?
- Plate and frame filter press
 - Shell and leaf filter press
 - Rotary drum filter
 - None of the above
- Q.8 For breaking of hard rock we use
- Crusher
 - Ball mill
 - Tube mill
 - Fluid energy mill
- Q.9 The operating speed of a ball mill should be _____ the critical speed.
- Less than
 - More than
 - Equal to
 - More than or equal to
- Q.10 Which of the following involves vibrations?
- Hammer mill
 - Ball Mill
 - Roll mill
 - Grizzly screen

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define sphericity.
- Q.12 Define the term size reduction.
- Q.13 State Bond's law of crushing.

- Q.14 Name any one type of jaw crusher.
- Q.15 Write name of any one grinder.
- Q.16 Mention any one type of vibrating screen.
- Q.17 What is mesh number?
- Q.18 Name one continuous filter.
- Q.19 What is sedimentation?
- Q.20 Name any one solid mixing equipment.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain particle shape and its importance in mechanical operations.
- Q.22 Draw a neat labeled diagram and explain fluid energy mill.
- Q.23 Differentiate between Jaw crusher and smooth roll crusher.
- Q.24 Derive the expression for specific surface of a particle mixture.
- Q.25 Explain Rittinger's law and its limitations.
- Q.26 Describe double arm kneader with the help of neat diagram.
- Q.27 Describe the construction and working of a Blake jaw crusher.
- Q.28 Explain the working principle and construction of a ball mill.